



Mansoura University
Faculty of Computers and Information
Department of Computer Science



Spring 2026

[SWE143] Software Project Management

Lecture 06: **Estimating : PERT**

Level 300: software engineering program

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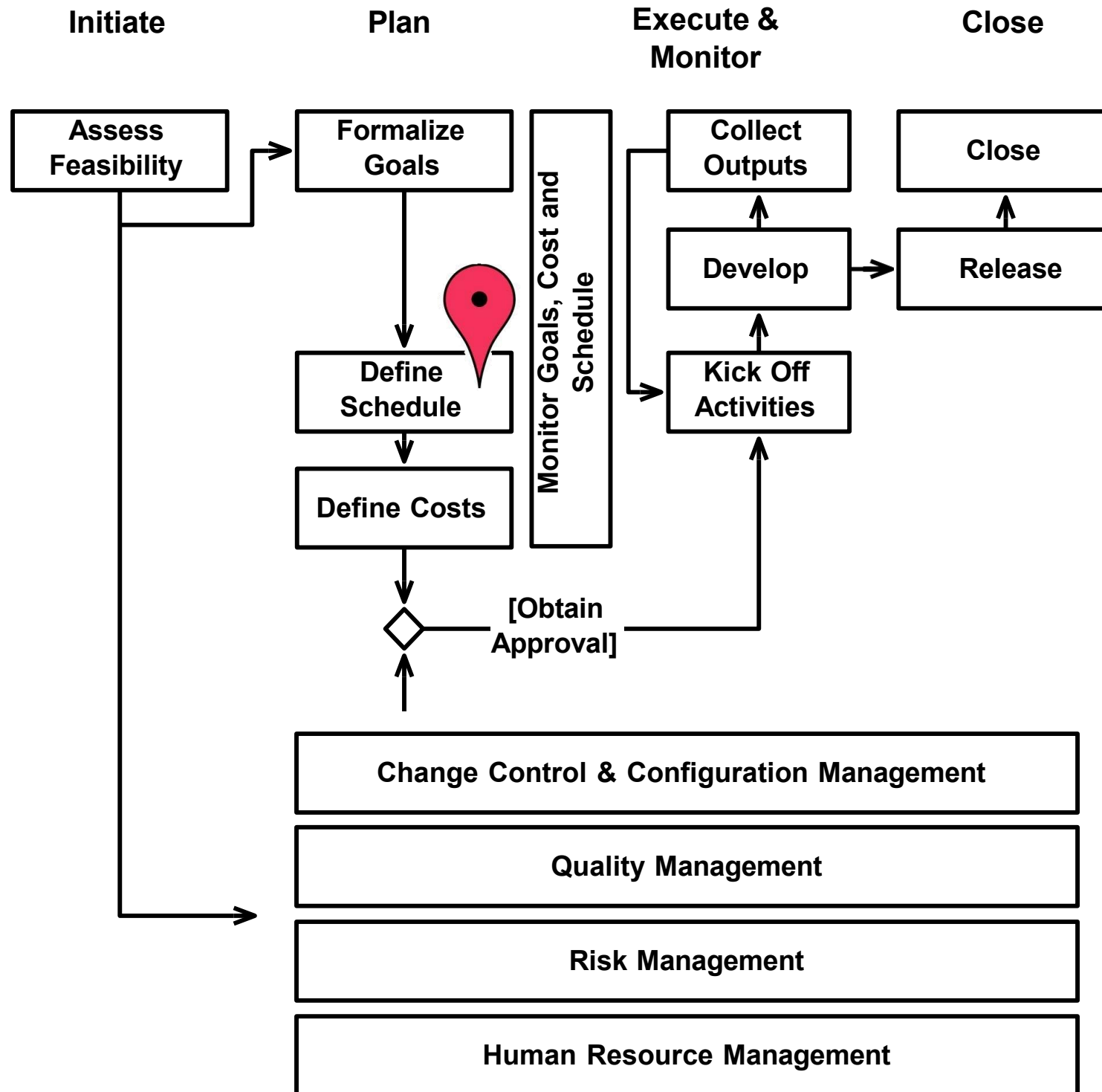
Estimating

A quote attributed to many people, from the Nobel prize-winning Quantum physicist Niels Bohr to legendary baseball player (and philosopher) Yogi Berra states: "It is difficult to make predictions, especially about the future."

Goals of the Unit

- Understand the fundamental (and simple) relation among Duration, Effort, and Manpower
- Understanding the perils مخاطر of estimations in software development
- Learn the techniques commonly adopted to estimate effort in projects

- Moving from the Work Breakdown Structure (WBS) to a Scheduled Plan.
- This step involves translating the hierarchical breakdown of tasks into a time-based project schedule, assigning durations and resources to each task.



Effort, Duration, and Resources

Estimation

Estimation is the process that determines the requirements to carry out an activity. These are expressed in terms of

- **Effort (Work):** how much work will the activity need to be completed
- **Resources:** type and quantity of resources available the activity
- **Duration:** how much time (how long) will the activity last for

Effort

- The **amount of work** an activity requires to be completed. A very good starting point.
- Measured in (work-)days, (work-)weeks, (work-)months
- Often the term **man-*** is also used (e.g. 3 man-months = 1 person working for 3 months; 3 people working for one month)
- Note: the work required in a project includes direct and indirect activities (i.e., getting the stuff done, but also email, communication, reports, meetings, ...)
 - for instance, an activity requiring 40 man-hours can be completed by a person working for 40 h

Resources

- The resources needed to carry the work out. Typically a constraints (limited)
- Expressed as **manpower**, that is, number of people (unit of work) and percentage of availability (presence of a resource)
- For instance: 1 person full time; 2 people at 50% part time.
- Types of resources:
 - Certain tasks might require **material resources** (e.g. bricks & pipes) or **equipment /tool** (e.g. a machine for DNA sequencing)
 - Material resources are consumed by the execution of an activity; equipment can be reused
- In software development usually resources = **manpower**
 - that is, the people involved in the project, like developers, testers, designers, project managers, etc.

Duration

- How long the activity will last for
- Measured in **hours, days, months, ...**
- Often:
 - 1 week = 5 days = 40 hours
 - 1 month (on average) = 4 weeks $\rightarrow 4 \times 5$ working days = 20 working days
- In some countries:
 - 1 week = 36 hours (7.12 hours/day)
 - considering 2 days of rest per week $\rightarrow 4 \times 8 + 1 \times 4$ or 4 days * 9 hours
- **Calendar time** differs from **duration**: calendar time includes non-working days, holidays, ...

A (simplistic) view

$$D = \frac{E}{M}$$

- Fix any two among D , E , and M (=manpower), and you get the third
- Typically, effort and manpower are the variables you will be working with (and derive **duration** from it)
- Notice also that **manpower** is

$$M = \sum_{i=1}^N p_i$$

N = number of resources

p_i = percentage of availability

Some Examples

- 1 week = 40 hours
- Effort: 40 man-hours; Resources: 1 @ 100% →
 $D = 40 \text{ man-hours} / 1 \text{ man} = 40 \text{ hours} = 1 \text{ week}$
- Effort: 80 hours; Resources: 2 @ 100% →
 $D = 80 \text{ man-hours} / 2 \text{ man} = 40 \text{ hours} = 1 \text{ week}$
- Effort: 80 hours; Resources: 1 @ 50% →
 $D = 80 / 50\% = 160 \text{ hours} = 4 \text{ weeks}$
(a person at 50% will be able to work 20 hours/week; it takes 4 weeks to get to the 80 hours needed for the activity)

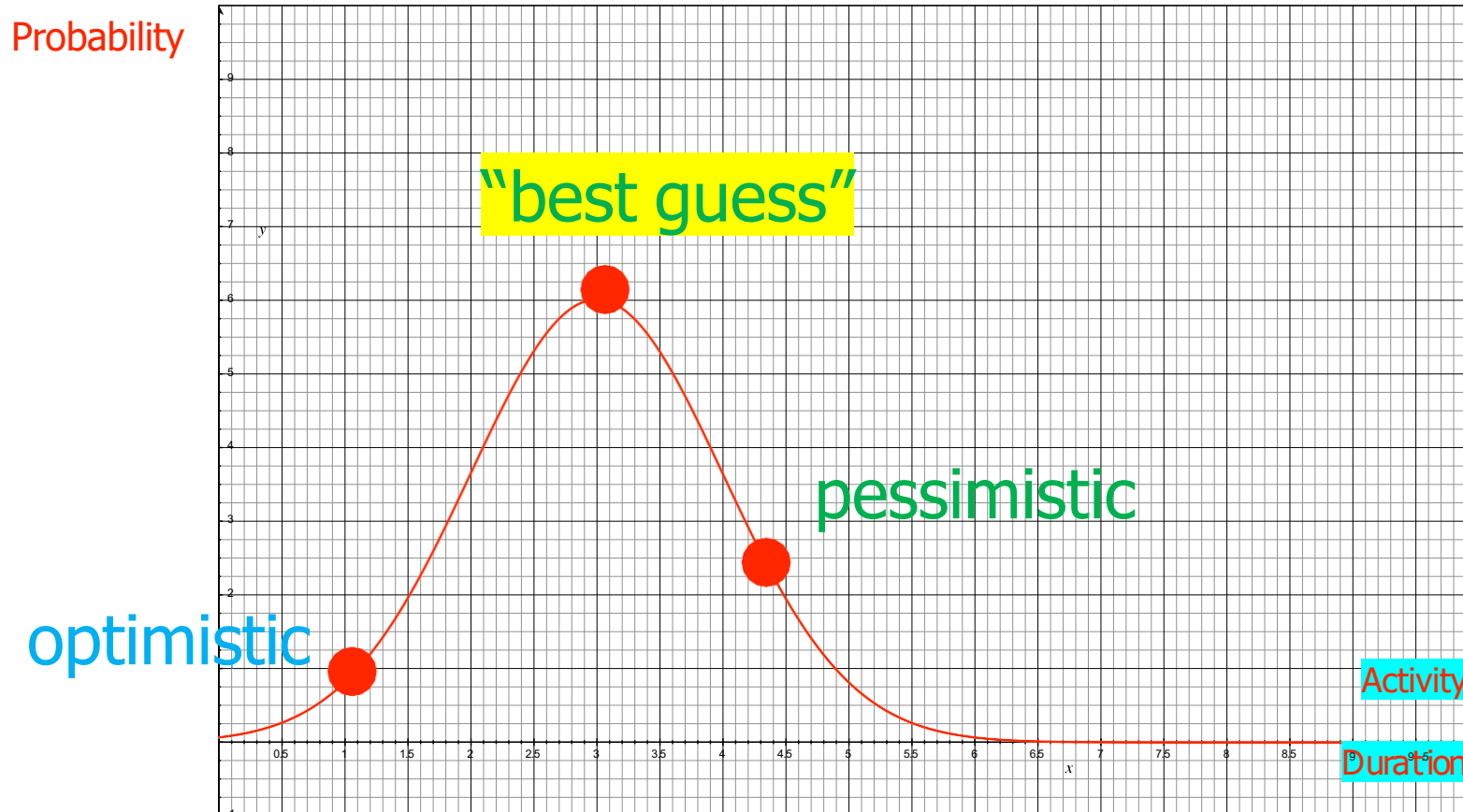
Important Remark

- The equation is a simplification... good enough for various cases (do not take it to extremes)
- The hypothesis of “take any two variables” in $D = E/M$ is not always reflected in practice (e.g. the variables are not completely independent)
 - For example, doubling manpower doesn’t always halve the duration — due to overhead, coordination costs, or task dependencies.
 - It's a core idea in **Brooks’ Law**, which says: *“Adding manpower to a late software project makes it later.”* Applies in other fields too.
- Estimating is hard: deciding how much effort a task requires is difficult ... in the next few lectures we will look at techniques and tools for estimation

Uncertainty in Planning

Uncertainty in planning

- Planning has a certain degree of uncertainty
- (In software and not only) we are over-optimistic: They end above the most likely duration
- “best guess” might also be a problem



Uncertainty in planning

- **Three** practices (not necessarily good) to account for uncertainty
 - **Implicit padding:** each activity includes some contingency time
 - **Explicit padding:** the contingency time is explicitly modeled as an activity
 - **React and re-plan:** when a delays occurs, you re-plan and re-define a new realistic schedule
- ❑ Contingency time is the amount of extra time that you add to your project schedule to account for possible delays, risks, or uncertainties.
- ❑ Some rules of thumb go up to doubling the value of the estimation.
- Some suggestions:
 - Always evaluate the cost of delays: Understand how delays might affect project timelines, budgets, and overall success.
 - Choose a strategy and make it clear (with yourself and with your stakeholders, if possible): Dilemma; our estimations do not satisfy upper management. **Expectation vs. reality mismatch**

Implicit Padding

- Each activity includes some extra duration/effort to take into account delays
- Estimations become inaccurate and difficult to control.
- Always assuming the worst-case scenario (being overly pessimistic) can create false certainty about the project's timeline and lead to misleading outcomes.
- Always being **pessimistic** (and always delivering earlier, according to a wrong pessimistic plan) is not necessarily good... the plan is still inaccurate
 - **Wasted Resources and Overhead:** the team might have slowed down or kept extra resources on standby, leading to unused capacity or idle time, as the task wasn't as difficult or risky as expected.
 - **Impact on Other Projects:** Interaction with other projects might still be a problem (you deliver earlier and the next project needs to re-allocate resources in order to start an activity earlier)

Explicit Padding

- The plan includes some extra activities or **slack** to take into account delays in finishing activities
- **Contingency** is not explicit in the plan. Data is accurate; no problems in budgeting/monitoring/...
 - Implicit contingencies are usually hidden within estimates and are not clearly visible to stakeholders.
 - Customers may feel that this additional buffer is unnecessary and could perceive it as padding — adding extra time or resources that aren't strictly needed.

React and Re-plan

- When a delay occurs, it is dealt with and specific actions are decided. The actions are incorporated into the plan, which is revised
- **Flexible:** takes into account different strategies for dealing with contingencies (e.g. removing dependencies, adding resources)
- This is not a **planning practice**. It is a monitoring and executing practice.
 - Monitoring is important in determining if resources and capacity are sufficient and well utilized and whether the project is moving as initially planned.
- **The plan** does not show possible alternative courses of actions to the occurrence of a risk/contingency

Estimation Techniques

Approaches to Estimation

- **Expert Judgement** is “quick and dirty” and based on experience. It can be applied either top-down (*estimate whole project first*) or bottom-up (*estimate tasks, then sum*).
- **PERT (Program Evaluation and Review Technique)** takes into account the probabilistic nature of estimations
- **Algorithmic Techniques** provide estimations by measuring specific qualities of a system and applying algorithms (Function Points, COCOMO, WebObjects)
 - **Function Points:** Measures software **functionality** (e.g., inputs, outputs) to estimate effort.
 - **COCOMO:** A formula-based model to estimate software effort, based on project **size** and complexity.
 - **WebObjects:** Estimates web app development effort by analyzing **technology**, UI complexity, and integration.

Expert Judgement

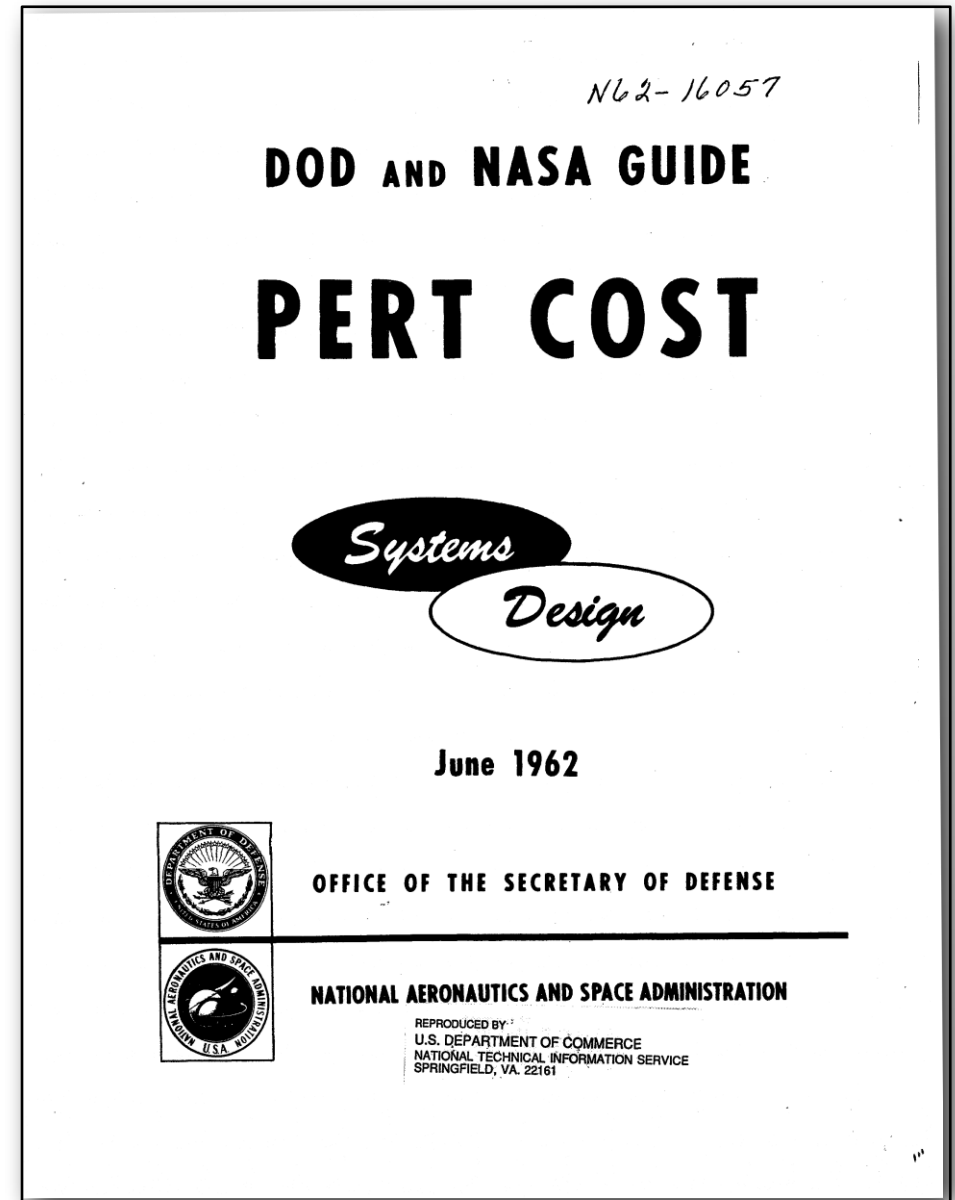
- Efficient and fast. Based on personal (rather than organizational) assets
 - provides estimations using experience or by looking at similar projects
- Underlying assumption: the project uses a product WBS
- Top-down
 - Start at the top of the WBS and break estimations as you move down
 - Tend to underestimate the duration or effort (since they might abstract away details)
- Bottom-up
 - Start at the bottom of the WBS (Leaves) and sum as you move up
 - Tend to overestimate the effort or the duration, because too much importance is given to details

PERT

Program Evaluation and
Review Technique

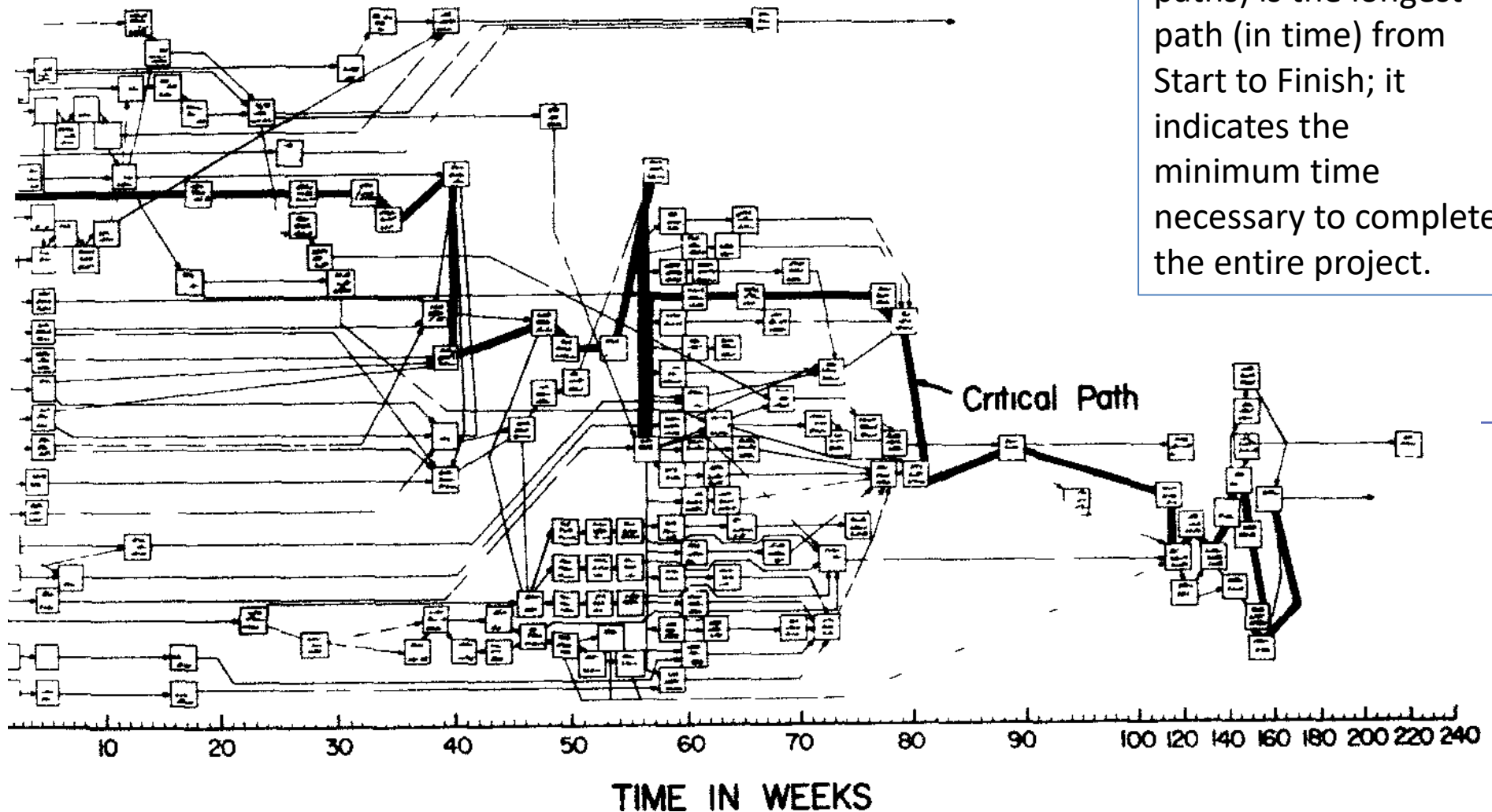
PERT

- Program Evaluation and Review Technique
- Developed in the sixties
- It is a **methodology** to define and control projects
- Variations exist (e.g. PERT/COST developed by NASA/DOD)
- Is a project management planning tool used to calculate the amount of time it will take to realistically finish a project.
 - The goal of the technique is to assess the probability of a plan to finish on a certain date, given the probabilistic estimations of the activities of composing the plan.



A Motivating Example

The critical path (or paths) is the longest path (in time) from Start to Finish; it indicates the minimum time necessary to complete the entire project.



PERT Formula

- Estimation in PERT is based on the idea that estimates are uncertain
 - Therefore uses duration ranges
 - And the probability of falling to a given range
- Uses an “expected value” (or weighted average) to determine durations

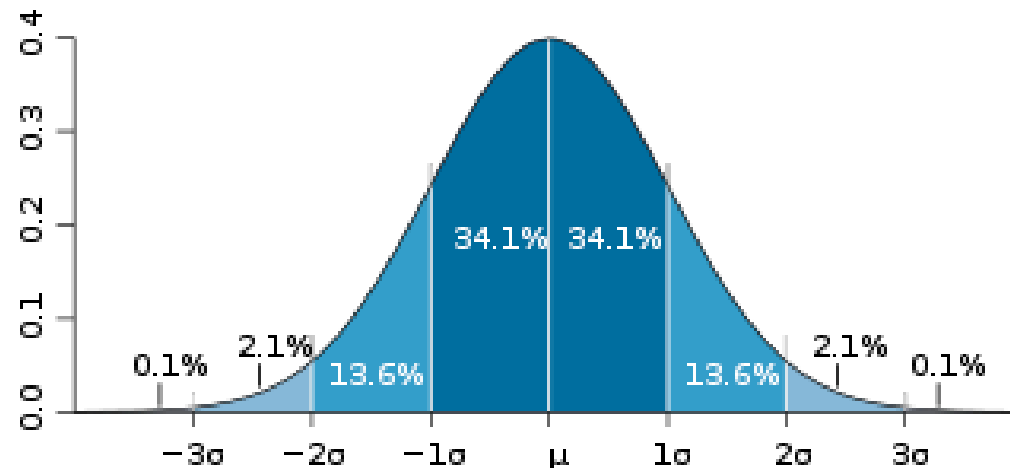
PERT

- For each task, three estimates:
 - The **optimistic** estimate of time is based on the assumption that an activity will not involve any difficulty during execution and it can be completed within a short period. **a or t_o**
 - A **pessimistic** estimate is made on the assumption that there would be unexpected problems during the execution of an activity and hence it would consume more time. **b or t_p**
 - The **most likely** time estimate is made in between the optimistic and the pessimistic estimates of time. **m or t_m**
 - Thus, the three estimates of time have the relationship

$$a < m < b$$

Variance and Standard Deviation

- **Variance (σ^2):** Measures the **spread** or **uncertainty** of an activity's duration estimate.
- **Standard deviation (σ):** The **square root of variance**. It shows how much the actual duration is likely to **deviate from the expected time**.
- **Example: normal distribution:** a bell shaped probability distribution function



Source: http://en.wikipedia.org/wiki/Normal_distribution

PERT Formula

- Task **duration** is a weighted average of three estimations:

$$t_e = \frac{(a + 4m + b)}{6}$$

$\text{Expected Time} = \frac{\text{Optimistic} + 4 \times \text{Most Likely} + \text{Pessimistic}}{6}$

t_e = expected time

a = optimistic time estimate

m = most likely time estimate

b = pessimistic time estimate

Measure Of Certainty

- The range for the time estimate is **b - a**
- Variance (σ^2) and standard deviation (σ) are given by:

$$\sigma^2 = \left(\frac{b-a}{6} \right)^2$$

$$\sigma = \left(\frac{b-a}{6} \right)$$

Where:

- **a** = Optimistic time
- **b** = Pessimistic time

PERT – Problem 1

- Two experts A and B examined an activity and arrived at the following time estimates. Determine which expert is more certain about his estimates of time.

Expert	Time Estimate		
	a	m	b
A	4	6	8
B	4	7	10

- Solution**

In the case of expert A, the variance = $\left(\frac{8-4}{6}\right)^2 = \frac{4}{9}$

As regards expert B, the variance = $\left(\frac{10-4}{6}\right)^2 = 1$

So, the **variance** is less in the case of **A**. Hence, it is concluded that the **expert A** is more certain about his estimates of time.

PERT – Problem 2

Find out the time required to complete the following project and the critical activities:

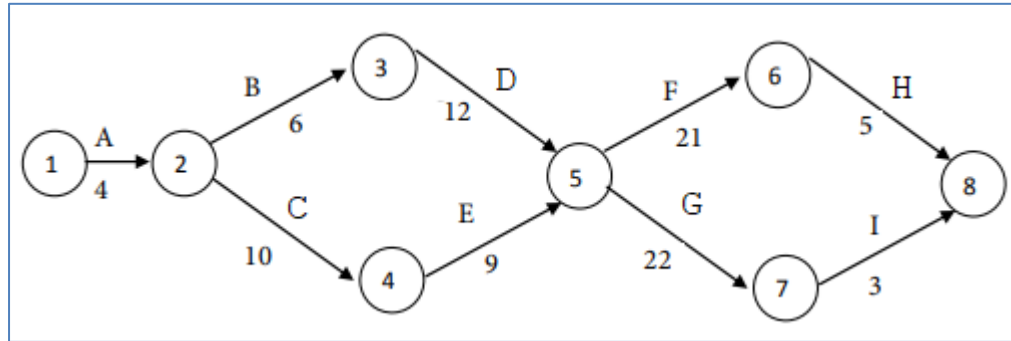
Solution: calculate time estimate for each activity

Activity	Predecessor Activity	Optimistic time estimate (to days)	Most likely time estimate (tm days)	Pessimistic time estimate (tp days)
A	-	2	4	6
B	A	3	6	9
C	A	8	10	12
D	B	9	12	15
E	C	8	9	10
F	D, E	16	21	26
G	D, E	19	22	25
H	F	2	5	8
I	G	1	3	5

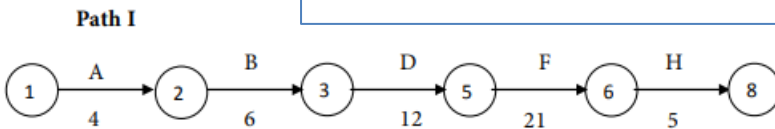
Activity	Optimistic time estimate (to)	4 x Most likely time estimate	Pessimistic time estimate (tp)	to + 4tm + tp	Time estimate $t_e = \frac{t_o + 4t_m + t_p}{6}$
A	2	16	6	24	4
B	3	24	9	36	6
C	8	40	12	60	10
D	9	48	15	72	12
E	8	36	10	54	9
F	16	84	26	126	21
G	19	88	25	132	22
H	2	20	8	30	5
I	1	12	5	18	3

PERT – Problem 2

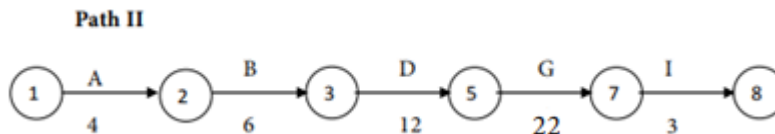
Solution: Using the single time estimates of the activities, we get the following network diagram for the project.



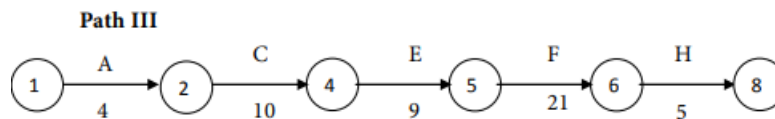
or utilizing CPM to find critical path by computing forward/backward passes



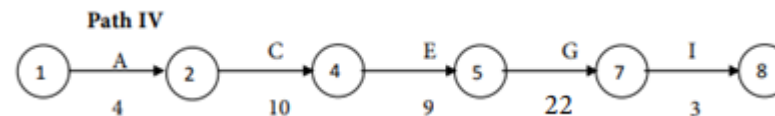
Time for the path: $4+6+12+21+5 = 48$ days.



Time for the path: $4+6+12+22+3 = 47$ days.



Time for the path: $4+10+9+21+5 = 49$ days.



Time for the path: $4+10+9+22+3 = 48$ days.

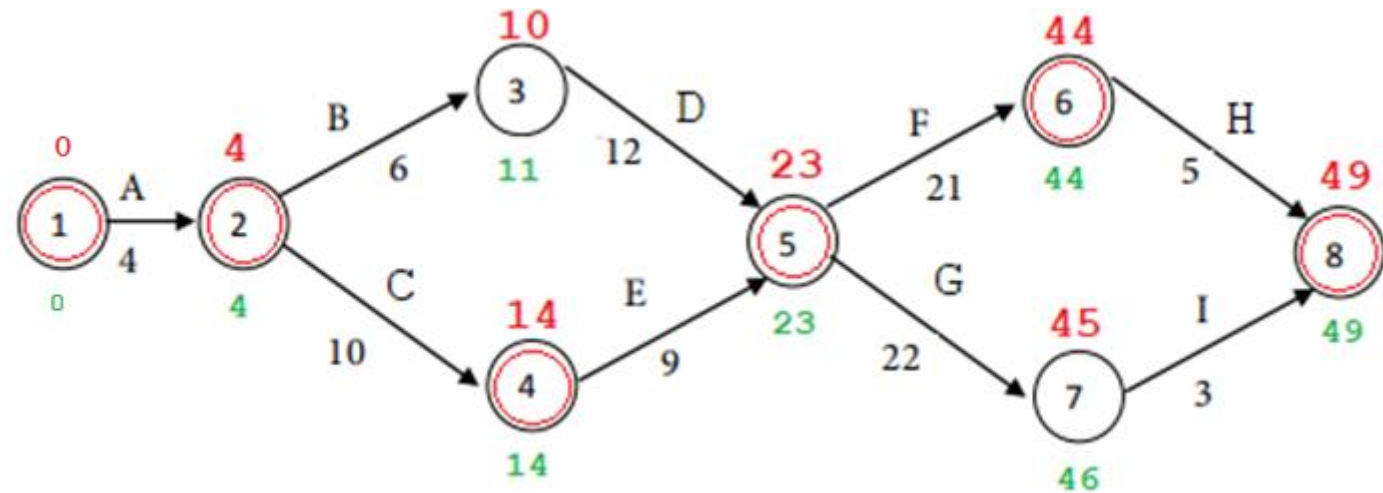
Compare the times for the four paths. **Maximum** of $\{48, 31, 49, 48\} = 49$. We see that Path III has the maximum time. Project time (Also called project length) = **49** days.

The critical activities are A, C, E, F and H.

The non-critical activities are B, D, G and I.

CPM

In the Activity-on-Arrow (AOA) method, activities are shown as arrows and nodes represent start and end events.



Critical Path Method steps

1. Perform the forward pass [Red]: compute **Earliest** Start date; $ES_j = \text{Max}(ES_i + D_{ij})$
 2. Perform the backward pass [Green]: compute **Latest** Finish date, $LF_i = \text{Min}(LF_j - D_{ij})$
 3. Compute the **slack**_{node} as the different between **ES** and **LF**. The critical path is made of all the tasks with zero slack
- In this simplified CPM, ES and EF (and LS and LF) still both exist — but only one of each pair is shown because the other is directly computed.
 - Every event node carries: **E = Earliest event time, L = Latest event time**
 - **Forward:** 1st node ES = 0, **Backward:** last node LF = node ES

Note: In the practical section, the **full (CPM)** using **Activity-on-Node (AON)** is used, where each activity has **four time** values: ES, EF, LS, and LF. **Slack** = LS – ES = LF – EF.

PERT – Problem 3

Find out the time, variance and standard deviation of the project with the following time estimates in weeks:

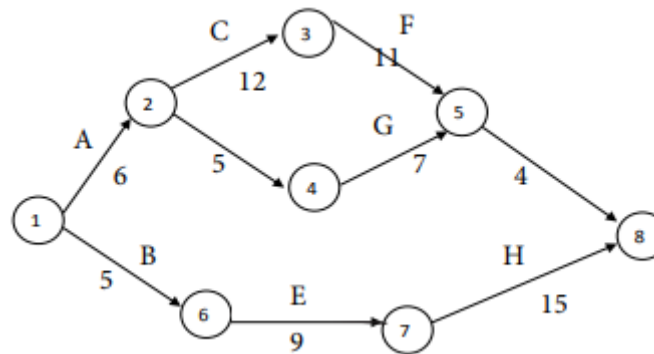
Solution: calculate time estimate for each activity

Activity	Optimistic time estimate (to)	Most likely time estimate (tm)	Pessimistic time estimate (tp)
1-2	3	6	9
1-6	2	5	8
2-3	6	12	18
2-4	4	5	6
3-5	8	11	14
4-5	3	7	11
6-7	3	9	15
5-8	2	4	6
7-8	8	16	18

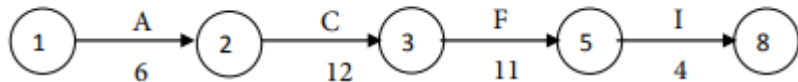
Activity	Optimistic time estimate (to)	4 x Most likely time estimate	Pessimistic time estimate (tp)	to+ 4tm + tp	Time estimate $t_e = \frac{t_o + 4t_m + t_p}{6}$
1-2	3	24	9	36	6
1-6	2	20	8	30	5
2-3	6	48	18	72	12
2-4	4	20	6	30	5
3-5	8	44	14	66	11
4-5	3	28	11	42	7
6-7	3	36	15	54	9
5-8	2	16	6	24	4
7-8	8	64	18	90	15

PERT – Problem 3

Solution: Using the single **time estimates** of the activities, we get the following network diagram for the project.



Path I



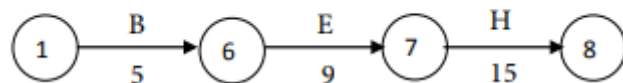
Time for the path: $6+12+11+4 = 33$ weeks.

Path II



Time for the path: $6+5+7+4 = 22$ weeks.

Path III



Time for the path: $5+9+15 = 29$ weeks

Compare the times for the three paths. **Maximum** of {33, 22, 29} = 33

Project time (Also called project length) = **33** weeks.

The critical activities are A, C, F and I.

The non-critical activities are B, D, G and H.

PERT – Problem 3

Solution: Calculation of Standard Deviation and Variance for the critical activities:

Critical Activity	Optimistic time estimate (to)	Most likely time estimate (tm)	Pessimistic time estimate (tp)	Range (tp - to)	Standard deviation = $\sigma = \frac{t_p - t_o}{6}$	Variance $\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$
A: 1 2	3	6	9	6	1	1
C: 2 3	6	12	18	12	2	4
F: 3 5	8	11	14	6	1	1
I: 5 8	2	4	6	4	2/3	4/9

- Sum of the variances for the critical activities = $1+4+1+ 4/9 = 58/9$ Weeks.
- Standard deviation of project time = $\sigma = \sqrt{\text{Variance}} = \sqrt{58/9} = \pm 2.54$ weeks
- The standard deviation (σ) of the project time is approximately 2.54 weeks, which represents the expected deviation from the expected project duration.

PERT – Problem 4

Using previous Problem-3,

Find the probability that the project will be completed in **36** weeks or less.

Solution:

The standard normal variate is given by the formula

$$Z = \frac{\text{Given value of } t - \text{Expected value of } t \text{ in the critical path}}{\text{SD for the critical path}}$$

$$\text{So, we got } z = \frac{36 - 33}{2.54} = 1.181$$

We refer to the Normal Probability Distribution Table.

Corresponding to $Z = 1.181$, we obtain the value of **0.881**

Therefore, the required probability is 0.88 i.e., There is **88%** chance that the project will be completed before 36 weeks. In other words, the chance that it will be delayed beyond 36 weeks is 12%

Z-score Calculation:

$$Z = \frac{x - \mu}{\sigma}$$

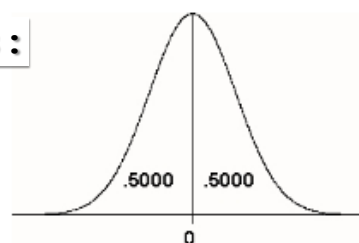
- value being examined x
- population mean μ
- population standard deviation σ

Standard Normal (Z) Table

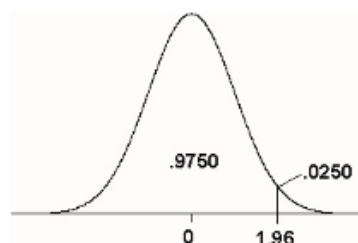
Values in the table represent areas under the curve to the left of Z quantiles along the margins.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

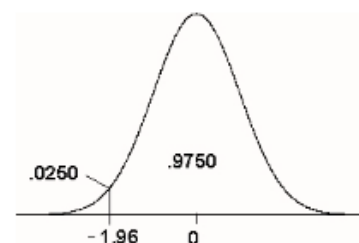
Examples:



$$P(z < 0) = 0.5000$$



$$P(z < 1.96) = 0.9750$$



$$P(z < -1.96) = 1 - 0.9750 = 0.0250$$

PERT – Problem 5

Using previous Problem-3,

Find the expected time to achieve the probability of 67%

Solution:

Given P = 0.67

Using Z table, we got Z= 0.44

$$Z = \frac{\text{expected time} - \text{mean time}}{SD}$$

$$Z = \frac{\text{expected time} - 33}{2.54} = 0.44$$

Expected time = 34.11 ~ 35 days

Thank You

